

Simple C Console Menu Program

A menu-driven C program that lets the user choose between a few small utilities: printing an ASCII table, calculating the total resistance of a parallel circuit, and finding the length of a string.

Features

- **A / B** - Print a full ASCII table (DEC, HEX, CHAR) for values 0–255
- **C** - Calculate total resistance of three resistors in parallel
- **D** - Calculate the length of a user-input string (custom function, no strlen)
- **X** - Exit the program

How It Works

The program runs in a loop, displaying a menu and reading a single character option via scanf. A switch statement handles each case:

- Options **A** and **B** both print the ASCII table (currently duplicated logic)
- Option **C** prompts for three resistor values (R1, R2, R3) and calculates the total resistance using the parallel resistance formula
- Option **D** reads a string and passes it to a custom stringlength() function that counts characters up to the null terminator
- Any invalid input shows an error message and re-displays the menu

$$R_{total} = 1 / (1/R1 + 1/R2 + 1/R3)$$

Building

```
gcc main.c -o menu_program
```

Running

```
./menu_program
```

Follow the on-screen menu and enter the corresponding capital letter (A, B, C, D, or X) when prompted.

Notes / Possible Improvements

- Options A and B currently perform the exact same action — consider removing the duplication or giving B a distinct purpose

- No input validation on resistor values (e.g. zero or negative numbers could cause division errors)
- `string_input` is a fixed-size buffer (100 chars) with no bounds checking on `scanf`